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AHA SCIENTIFIC STATEMENT

Core Components of Cardiac Rehabilitation Programs: 2024 Update: A Scientific Statement From the American Heart Association and the American Association of Cardiovascular and Pulmonary Rehabilitation

Endorsed by the American College of Cardiology

Todd M. Brown, MD, MSPH, FAHA, Chair; Quinn R. Pack, MD, MSc, FAHA, Vice Chair; Ellen Aberegg, MA, LD, RD; LaPrincess C. Brewer, MD, MPH, FAHA; Yvonne R. Ford, PhD; Daniel E. Forman, MD, FAHA; Emily C. Gathright, PhD; Sherrie Khadanga, MD; Cemal Ozemek, PhD, ACSM-CEP; Randal J. Thomas, MD, MS, FAHA; on behalf of the American Heart Association Exercise, Cardiac Rehabilitation and Secondary Prevention Committee of the Council on Clinical Cardiology; Council on Cardiovascular and Stroke Nursing; Council on Lifestyle and Cardiometabolic Health; and Council on Quality of Care and Outcomes Research

ABSTRACT: The science of cardiac rehabilitation and the secondary prevention of cardiovascular disease has progressed substantially since the most recent American Heart Association and American Association of Cardiovascular and Pulmonary Rehabilitation update on the core components of cardiac rehabilitation and secondary prevention programs was published in 2007. In addition, the advent of new care models, including virtual and remote delivery of cardiac rehabilitation services, has expanded the ways that cardiac rehabilitation programs can reach patients. In this scientific statement, we update the scientific basis of the core components of patient assessment, nutritional counseling, weight management and body composition, cardiovascular disease and risk factor management, psychosocial management, aerobic exercise training, strength training, and physical activity counseling. In addition, in recognition that high-quality cardiac rehabilitation programs regularly monitor their processes and outcomes and engage in an ongoing process of quality improvement, we introduce a new core component of program quality. High-quality program performance will be essential to improve widely documented low enrollment and adherence rates and reduce health disparities in cardiac rehabilitation access.

Key Words: AHA Scientific Statements ■ cardiac rehabilitation ■ cardiovascular diseases ■ secondary prevention

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Cardiac rehabilitation (CR) reduces mortality rates and improves the quality of life of patients with cardiovascular disease (CVD),[1] and its use is supported by American College of Cardiology/American Heart Association performance measures and numerous clinical practice guidelines.[2–7] The structure of a CR program was defined in the Social Security Act[8] and is summarized in Table 1. CR programs can be located in a hospital (on or off the main hospital campus) or physician's office, must have a physician medical director, and must be supervised by a physician or nonphysician practitioner (eg, physician assistant, nurse practitioner, clinical nurse specialist) with appropriate expertise in CVD pathophysiology who is immediately available to provide consultation or handle medical emergencies at all times of CR service provision. Programs must include physician prescribed exercise, cardiac risk factor modification, psychosocial assessment, and outcomes assessment. Each

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Table 1. Cardiac Rehabilitation Program Requirements as Defined in the Social Security Act[8]

Location
Hospital (on or off campus) or physician's office
Physician or nonphysician practitioner supervision
Individualized treatment plan
Signed by a physician and updated every 30 d
Qualifying diagnoses
Acute myocardial infarction in the past 12 mo
Coronary artery bypass graft surgery
Coronary artery angioplasty or stenting
Heart valve repair or replacement
Heart or heart–lung transplantation
Stable angina
Stable chronic heart failure*
Required components
Physician-prescribed exercise
Cardiac risk factor modification
Psychosocial assessment
Outcomes assessment

*Defined as having a left ventricular ejection fraction ≤35% and New York Heart Association class II to IV symptoms despite being on optimal heart failure therapy for ≥6 wk.

patient must have an individualized treatment plan (ITP) signed by a physician and updated every 30 days. There are additional requirements for qualifying diagnoses, session number and duration, and intensive CR programs. As a result, CR programs consist of a multidisciplinary team of clinicians, who may include physicians, nurses, clinical exercise physiologists, behavioral health experts, dietitians, physical and respiratory therapists, and others, who collaborate to deliver these services.

CR historically has been provided almost exclusively in person in a hospital or physician's office setting. However, CR programs and patients are increasingly seeking new methods to deliver CR sessions.[9] These new care models have been summarized by the Million Hearts Cardiac Rehabilitation Think Tank.[10] Virtual (synchronous) sessions are conducted by clinicians working in CR programs who interact with patients for the entire duration of the session using real-time audiovisual communications technology; during remote (asynchronous) sessions, there is no real-time interaction.[10] Programs do not have to provide only 1 form of delivery. A patient may receive a hybrid of these delivery models, with varying combinations of in-person, virtual (synchronous), or remote (asynchronous) sessions while undergoing CR.[11]

These modes of CR delivery conceptually can overcome several barriers to CR participation, including geographic inaccessibility, transportation, and competing priorities or demands, thereby enhancing CR uptake and adherence.[12–15] There is evidence to suggest that these alternative delivery models have similar efficacy and safety as traditional in-person CR in improving CVD risk factors, mortality risk, and health-related quality of life, at least in patients at low to moderate risk.[12,13,16,17] Furthermore, hybrid models have the potential to bolster equity in CR participation by offering multiple options to meet the needs of diverse populations.[18]

However, the adoption of novel models of delivery must not change the fundamental therapeutic foundation in what is delivered. CR is a medically supervised secondary prevention program with a physician medical director, day-to-day medical supervision by a physician or nonphysician practitioner, and core components, all of which must be included for a program to be classified as CR. Alternative delivery models that do not meet these statutorily defined standards are not CR and should not be viewed as an acceptable alternative to CR.

To provide guidance to programs and ensure high-quality care, the American Association of Cardiovascular and Pulmonary Rehabilitation (AACVPR) and American Heart Association have defined the core components of CR and secondary prevention programs.[19] This article updates these core components and addresses new methods to deliver these services (Figure). Details regarding the evaluation, intervention, and goals of each core component are provided in Table 2, and the scientific evidence is summarized in the Supplemental Table. We recognize that not every program will be able to, or choose to, implement every bullet point listed in Table 2. Rather, we aim to summarize the scientific evidence for each aspect to facilitate informed and scientifically sound decisions by CR programs and medical directors about the best and most effective ways to implement these core components in individual CR programs. We have summarized the major updates in this scientific statement in Table 3 and display how these changes fit within the federal statute, clinical guidelines, and performance measures relevant to CR.

PATIENT ASSESSMENT

CR aims to address medical, physical, and behavioral aspects of heart disease comprehensively to foster maximal recovery, risk reduction, and sustainable improvements in health. Vulnerability and progression of atherosclerosis, heart failure, arrhythmia, and other CVDs are often contingent on CVD risk factors, comorbidities, frailty, lifestyle (eg, physical activity, nutrition), medication adherence, substance use or misuse, psychological distress, and social determinants of health. Thus, patient assessment demands a comprehensive overview of health and lifestyle habits, extending well beyond the primary cardiac diagnosis.

A medical history and physical examination are elemental (Table 2). Relevant concerns include current and previous medical and surgical diagnoses and

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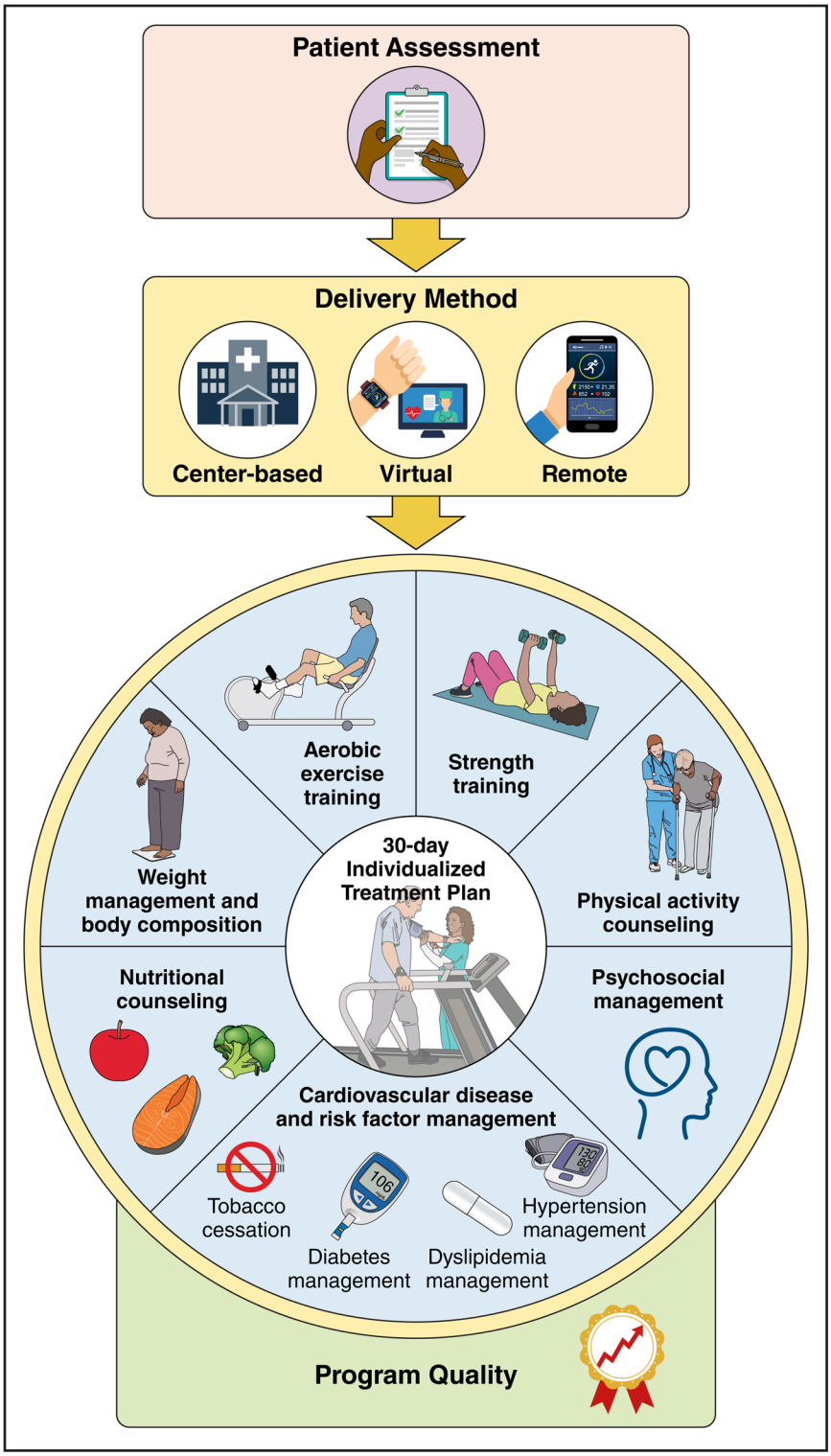

Figure. Delivery methods and core components of cardiac rehabilitation programs.
The individual core components of cardiac rehabilitation programs surround the central component of patient assessment and rest on a foundation of high program quality. These components can be delivered in person, virtually (using synchronous audiovisual communications technology), remotely (asynchronously with no real-time interaction), or using a hybrid of these methods to meet the individual needs of the patient.

procedures, cardiovascular function, traditional atherosclerotic CVD risk factors, comorbid disease factors that affect cardiovascular health, medical issues that may affect CR and lifestyle management goals, history of falls or syncope, symptoms, and all prescription and nonprescription medications, including supplements and vaccinations.

Social history (Table 2) is similarly fundamental. Review should include medications and medical care, including affordability and administration; physical activity; shopping and food patterns; alcohol, tobacco, or drug dependencies; sleep patterns; marital status; living situation; social supports and routines; health literacy; education and socioeconomic status; employment, personal

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Table 2. Core Components of Cardiac Rehabilitation Programs

Component	Details
Patient assessment	
Evaluation	Medical history
	Cardiovascular medical diagnoses and surgical procedures (eg, coronary artery disease, heart failure, arrhythmias)
	Cardiovascular function (eg, ventricular function, structural or hemodynamic abnormalities, residual ischemic burden)
	Atherosclerotic CVD risk factors (eg, hypertension, hyperlipidemia, elevated glucose level, elevated hemoglobin A1c level, tobacco use, family history, increased body mass index, abdominal obesity, poor diet, psychosocial stress, physical inactivity, low cardiorespiratory fitness)
	Comorbid conditions associated with increased CVD risk (eg, diabetes, hypertension, peripheral arterial disease, cerebrovascular disease, chronic kidney disease, obstructive sleep apnea, pulmonary hypertension, congenital heart disease)
	Other comorbid conditions (eg, pulmonary disease, gastrointestinal disease, musculoskeletal or neuromuscular disorders, cancer, depression, human immunodeficiency virus infection, cognitive impairment/confusion, falls, sensory impairments, motor or gait impairments, psychiatric disease, pain)
	Cardiovascular and noncardiovascular symptoms
	Prescription and nonprescription medications and supplements, including dose, frequency, and adherence
	Vaccination status (eg, influenza, pneumovax, COVID-19)
	Social history
	Medication access, adherence, and affordability
	Physical activity patterns (extent of daily activity, sedentariness, and intentional exercise)
	Sleep patterns (sleep quantity and quality)
	Substance use or misuse (eg, alcohol, tobacco, vaping, marijuana, cocaine, opioids)
	Alcohol, tobacco, and other substance use (eg, vaping, marijuana, cocaine, opioids)
	Marital status and living/home situation (alone, with family, other)
	Social support structure (eg, spouse/partner, children, other family or community members)
	Shopping and food patterns (eg, who does shopping and cooking, frequency of restaurant dining)
	Health literacy and education
	Work or employment routines
	Goals of care and perceived barriers to care
	Personal preferences of care
	Physical examination
	Vital signs (including blood pressure, heart rate and regularity)
	Cardiac and pulmonary examination (including auscultation of heart and lungs, palpation and inspection of lower extremities for edema and presence of arterial pulses, cardiovascular procedure wound sites)
	Mental status and mood
	Cognitive function
	Physical function (eg, station, gait, and balance)
	Frailty
	Fall risk
	12-lead ECG
	Baseline telemetry and changes with first exercise session
	Functional assessment (see aerobic exercise component)
	Home environment
	Location and characteristics of dwelling (eg, house, apartment, presence of stairs)
	Family support
	Transportation and associated logistics
	Access to safe space for exercise either at home or in the community
	Access to healthful foods
Intervention	Individualized treatment plan
Goals/outcomes	Final outcomes report

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Table 2. Continued

Nutritional counseling	
Evaluation	Use validated diet assessment tool (food frequency questionnaire) that reflects current guidelines for preventative cardiovascular health Review results of questionnaire and solicit patient description of typical eating habits (eg, home, out, fast, distracted, alone) Perform assessment at the beginning and end of CR and relate to relevant clinical improvements (eg, weight loss, improved hemoglobin A1C level)
Intervention	A CR staff member will review results with the participant, and, using recommended measures or those established by the program, refer the participant to a dietitian, educational programming, or individualized goal setting Individualized goal setting is initiated early in the CR program, including family members if possible Progress toward goals achievement is reviewed weekly, formally or informally (if the goal is not met, the goal is reset to make it achievable; if the goal is met, the CR staff member will discuss rewards, review diet priorities, and set next goal) Educational classes should be encouraged for factual basis for recommendations and interaction with fellow participants; live or video demonstrations of food preparation, grocery decision-making, and portion assessment are important
Goals/outcomes	Nutrition-related behavioral goals established at program entry and progress evaluated weekly or monthly Goals and progress documented on the individualized treatment plan and medical record Expected outcomes: diet scores move toward recommended; improved relevant clinical measures; demonstrated or verbalized ability to make healthy food decisions independently
Weight management and body composition	
Evaluation	Measurements of weight and body mass index are simple and feasible measurement methods, but lack the ability to differentiate body fat and lean body mass Skinfold measurements and bioelectrical impedance lack precision and accuracy in assessing body fat and lean body mass Waist measurements (waist circumference, waist-to-hip ratio) are low-cost, highly feasible, and moderately accurate assessments of body fat Advanced methods (dual X-ray absorptiometry, magnetic resonance imaging) are more accurate and precise in measuring body fat and lean body mass but have higher costs and lower feasibility compared with other methods
Intervention	Lifestyle therapy, including dietary and exercise therapies, can reduce body fat, increase lean body mass, and reduce body weight High-intensity interval training may help improve body composition more than moderate-intensity continuous training Dietary supplements and medications may help improve weight control and body composition Surgical or interventional therapies can reduce body weight
Goals/outcomes	Primary goal: improvement in body composition by decreasing body fat or increasing lean body mass Secondary goal: optimal levels of body composition
CVD and risk factor management: hypertension	
Evaluation	Measure ≥2 seated resting blood pressures on ≥2 visits after having the patient relax, sitting in a chair (feet on floor, back supported, legs uncrossed, not talking) for >5 min Measure blood pressure in both arms at program entry Assess current treatment and medication adherence Assess use of nonprescription drugs that may adversely affect blood pressure
Intervention	Provide or monitor antihypertensive therapy in concert with primary clinician as follows: Recommend lifestyle modifications, including regular physical activity or exercise, weight management, dietary sodium restriction, enhanced dietary potassium intake, and increased consumption of fresh fruits, vegetables, and low-fat dairy products; moderation or abstinence of alcohol; and smoking cessation Report consistently elevated blood pressure to referring physicians and support patients in monitoring response Partner with program medical director and other clinicians (eg, hypertension specialists) through care-coordinated pathways to provide antihypertensive medications according to guideline recommendations. Use of physician-approved medication algorithms may support the safe and efficient provision of medications to patients in staff-limited programs Educate patients on the importance of medication adherence and regularly assess medication adherence during CR sessions
Goals/outcomes	SBP <130 mm Hg and DBP <80 mm Hg
CVD and risk factor management: dyslipidemia	
Evaluation	Obtain a fasting lipid profile if not recently obtained Assess treatment and medication adherence Assess use of over-the-counter supplements

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Table 2. Continued

Intervention	Provide or monitor lipid-lowering therapy in concert with primary clinician as follows: Recommend lifestyle modifications, including regular physical activity or exercise, weight management, and dietary counseling to consume a diet that emphasizes the intake of vegetables, fruits, whole grains, low-fat dairy products, fish, legumes, nontropical vegetable oils, and nuts, and limits the intake of sweets, sugar-sweetened beverages, and red meats Report elevated blood lipid values to referring physicians Partner with program medical director and other clinicians (eg, lipids specialists) through care-coordinated pathways to provide lipid-lowering medications according to guideline recommendations. Use of physician-approved medication algorithms may support the safe and efficient provision of medications to patients in staff-limited programs. Educate patients on the importance of medication adherence and regularly assess medication adherence during CR sessions
Goals/outcomes	Patients with atherosclerotic CVD should have a low-density lipoprotein cholesterol level <70 mg/dL and <55 mg/dL if considered to be at very high risk
CVD and risk factor management: diabetes	
Evaluation	Obtain fasting blood glucose and hemoglobin A1C levels if not recently obtained Assess treatment and medication adherence
Intervention	Provide or monitor diabetes therapy in concert with primary clinician as follows: Recommend lifestyle modifications, including regular physical activity or exercise, weight management, and dietary counseling to reduce hemoglobin A1C values, such as a Mediterranean, Paleolithic, low carbohydrate, high-protein, vegetarian, or nut-enriched diet Report elevated blood glucose values to referring physicians Partner with program medical director and other clinicians (eg, endocrinology teams) through care-coordinated pathways to provide diabetes medications according to guideline recommendations. Use of physician-approved medication algorithms may support the safe and efficient provision of medications to patients in staff-limited programs Educate patients on the importance of medication adherence and regularly assess medication adherence during CR sessions
Goals/outcomes	Hemoglobin A1C <7% for most patients
CVD and risk factor management: tobacco cessation	
Evaluation	Determine smoking status at program entry, including documenting previous use and quit date, if applicable Determine the patient's readiness to change if currently smoking Assess for current treatment and adherence in those who have already begun smoking cessation therapy before enrollment
Intervention	Provide or monitor smoking cessation interventions in concert with primary clinician as follows: Provide counseling on the importance and health benefits of smoking cessation Refer interested patients to dedicated smoking cessation programs Partner with program medical director and other clinicians (eg, nicotine dependency counselors) through care-coordinated pathways to provide smoking cessation medications according to guideline recommendations. Use of physician-approved medication algorithms may support the safe and efficient provision of medications to patients in staff-limited programs
Goals/outcomes	All current and recent smokers should receive some form of smoking cessation intervention All current and recent smokers ideally would be abstaining from smoking by program completion
Psychosocial management	
Evaluation	Assess using validated screening tools or interview of psychological distress, including depression, perceived stress, and anxiety Other relevant indicators to assess include sexual dysfunction, anger, loneliness, social isolation, and problematic substance misuse or substance use patterns that interfere with CR Evaluation should identify use of external psychological or psychiatric management
Intervention	Individual or group education and counseling on psychological adjustment to CVD, health behavior or lifestyle change, stress management, relaxation training, self-care, sex and intimacy, and social support Curation of a supportive CR environment and leveraging community resources Patients with distress beyond the scope of CR management should be referred for specialized evaluation and treatment Technology access and remote psychiatric safety should be considered in virtual and remote delivery of psychosocial care
Goals/outcomes	Absence of clinically elevated psychological distress and interference of other psychosocial concerns with management of cardiovascular health Sustained health behavior change, stress management, and social support When indicated, a tailored plan for persisting psychosocial challenges should include self-care strategies, medical team coordination, community resources, or external referrals to ongoing psychosocial management

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Table 2. Continued

Aerobic exercise training	
Evaluation	Graded exercise test The treadmill or cycle exercise test protocol should be individually selected on the basis of the patient's predicted cardiorespiratory fitness, with the goal of reaching maximal or near maximal exertion within 8 to 12 min
	6-min walk test Patients asked to walk as far as they can within the 6-min time period Ideally recommended to be performed in a corridor that allows for a 30-meter walking course; previous studies, however, have performed this test on 20- to 50-meter courses
Intervention	Frequency 3–5 d/wk
	Intensity Moderate 40%–59% of heart rate reserve Rating of perceived exertion of 12 or 13 on 6–20 Borg Scale Vigorous 60%–89% of heart rate reserve Rating of perceived exertion of 14–17 on 6–20 Borg Scale 10 beats/min below heart rate associated with any of the following criteria: Onset of angina or other symptoms of cardiovascular insufficiency Plateau or decrease in SBP, SBP >240 mm Hg, or DBP >110 mm Hg >1 mm or horizontal or downsloping ST-segment depression Radionuclide evidence of reversible myocardial ischemia or echocardiographic evidence of moderate to severe wall motion abnormalities during exertion Increased frequency of ventricular dysrhythmias Other substantial electrocardiographic disturbances (eg, second- or third-degree atrioventricular block, atrial fibrillation, supraventricular tachycardia, complex ventricular ectopy) Other signs or symptoms of exertional intolerance
	Time 20–60 min, including warm-up and cool-down
	Type Treadmill, cycling, elliptical, rowing, stair climbing, arm/leg ergometry, recumbent stepper, and other modalities allowing for continuous or interval aerobic training
	Volume Assess volume (frequency×intensity×time) completed per session and each week Can be expressed in calories or MET×min of activity Progression Increase 1 component of frequency intensity time type at a time Increase duration by 1–5 min per session until the goal duration has been achieved Increase intensity after goal duration has been achieved A 5%–10% increase in intensity is generally well tolerated
Goals/outcomes	Increase peak oxygen uptake by ≥15% Increase estimated exercise session peak metabolic equivalent by ≥40% from third to last session Increase 6-min walk distance by ≥10%
Strength training	
Evaluation	One or multiple repetition maximum tests as follows: Have patient perform a familiarization session with the respective equipment and movements The patient performs a warm-up consisting of multiple submaximal repetitions of the exercise that will be used to determine the 1 or multiple repetition maximum Determine the 1 or multiple repetition maximum within 4 trials with rest periods of 3 to 5 min between trials Select an initial weight that is within the individual's perceived capacity (≈50%–70% of capacity) Resistance is progressively increased by 5%–10% for upper-body or 10%–20% for lower-body exercise from the previous successful attempt until the individual cannot complete the selected repetitions; all repetitions should be performed at the same speed of movement and range of motion to instill consistency between trials The final weight lifted successfully is recorded as the absolute 1 repetition maximum or multiple repetition maximums

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Table 2. Continued

Intervention	Frequency
	2–3 nonconsecutive d/wk
	Intensity
	10–15 repetitions
	40%–60% of 1 repetition maximum
	Rating of perceived exertion of 11–13 on 6–20 Borg scale
	Time
	1–3 sets; 8–10 different exercises focused on major muscle groups
	Type
	Equipment that is safe and comfortable for the individual to use
Goals/outcomes	Prevent/treat frailty and reduce risk of falling
	The following assessment tools can be used to identify patients at risk for frailty or falls:
	Timed Up and Go
	Patient rises from a chair, walks 3 min at normal pace, turns around, returns to chair, and sits down
	An older adult who takes ≥12 s to perform task is at risk for falling
	30-Second Chair Stand
	With hands placed on the opposite shoulder, crossed, and feet flat on the floor, the patient fully stands and sits back down as many times as possible within 30 s
	Scores considered below average are as follows:

Age	Men	Women
60–64	<14	<12
65–69	<12	<11
70–74	<12	<10
75–79	<11	<10
80–84	<10	<9
85–89	<8	<8
90–94	<7	<4

	4-Stage Balance Test
	This test consists of 4 standing positions that get progressively more difficult to maintain. The patient is asked to hold a position for 10 s without moving their feet or needing support. If they are not able to, the test stops.
	1. Feet side by side
	2. Instep of 1 foot touching the big toe of the other foot
	3. Tandem stance where 1 foot is in front of the other, heel touching toe
	4. Stand on 1 foot
	Berg Balance Scale
	14-item scale that includes multiple balance changes, including position change, transferring, standing balancing tests, reaching, turning 360°, and retrieving object from the floor. Each item is scored from 0 to 4 and the scores are summed for a total score of 0–56. Higher scores indicate better performance.
	41–56: independent
	21–40: walking with assistance
	0–20: wheelchair-bound
Physical activity counseling	
Evaluation	Assessment of physical activity levels (using self-report, pedometers, or other activity monitors)
Intervention	Education and counseling on the cardiovascular and general health benefits of regular physical activity
Goals/outcomes	≥150 min/wk moderate-intensity activity or 75 min vigorous activity
Program quality	
Evaluation	Assessment of referral, enrollment, and adherence is essential for system-level quality improvement in CR
	Assessment of patient outcomes and risk factors is essential for patient-level quality improvement in CR
	Assessment is carried out before, during, and after interventions to assess effects on the quality of CR delivery
	Various methods can be used for evaluation, including article-based methods, electronic medical record tools, the American Association of Cardiovascular and Pulmonary Rehabilitation national registry, or other administrative database methods
	Evaluation methods should identify barriers and potential solutions for quality improvement, especially in subgroups within which CR participation is very low (eg, heart failure, stable angina, peripheral arterial disease, women, people of underrepresented races and ethnicities, non-English language speakers)

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Table 2. Continued

Intervention	Interventions have been found to improve CR referral, enrollment, and adherence CR participation in underserved populations appears to increase with the same tools
Goals/outcomes	Improvement in CR referral, enrollment, and adherence Improvement in clinical outcomes (eg, risk factor control, CVD morbidity and mortality events, psychological health) An outcomes report should be entered into the medical record and forwarded to the referring physician for review

CR indicates cardiac rehabilitation; CVD, cardiovascular disease; DBP, diastolic blood pressure; and SBP, systolic blood pressure.

challenges, and perceived barriers to care; goals of care; and personal preferences.

A resting 12-lead ECG is indicated if a recent tracing is not available in the medical record. Functional assessment is also indicated to guide the exercise prescription. Aerobic, strength, and balance domains are all germane,[28] and are described in more detail in subsequent sections.

In addition to standard physical fitness assessments, several additional tests can be used to enhance functional assessment for adults who are relatively more frail, deconditioned, or relatively weakened by their disease.[29,30] For these patients, CR provides important opportunities to regain sufficient function and resilience to maintain independence. The 5 Times Sit-to-Stand[31] is an example of an assessment that can be applied

Table 3. Summary of Changes in the 2024 Scientific Statement on the Core Components of Cardiac Rehabilitation Programs in the Context of the Federal Statute, Previous Scientific Statement, and Applicable Guidelines or Performance Measures

SSA defined components[8]	2007 Core components[19]	2024 Core components	Major changes in the 2024 scientific statement	Associated guidelines or performance measures
Individualized treatment plan	Patient assessment	Patient assessment		
Cardiac risk factor modification	Nutritional counseling	Nutritional counseling		AHA/ACC Lifestyle Management Guideline[20]
	Weight management	Weight management and body composition	Added assessment of body composition in addition to simply the assessment of weight	Multi-Society Secondary Prevention of ASCVD Guideline[21]
	Blood pressure management	Blood pressure management		Multi-Society Secondary Prevention of ASCVD Guideline[21] Muti-Society Blood Pressure Guideline[22]
	Lipid management	Dyslipidemia management		Multi-Society Secondary Prevention of ASCVD Guideline[21] Multi-Society Blood Cholesterol Guideline[23]
	Diabetes management	Diabetes management		Multi-Society Secondary Prevention of ASCVD Guideline[21] ADA Diabetes Management Guideline[24]
	Tobacco cessation	Tobacco cessation		Multi-Society Secondary Prevention of ASCVD Guideline[21] ATS Guideline on Tobacco Cessation[25]
Psychosocial assessment	Psychosocial management	Psychosocial management		
Physician prescribed exercise	Exercise training	Aerobic exercise and strength training	Separated strength training and aerobic exercise training as separate components to reflect the interval evidence base for strength training since the 2007 publication	
	Physical activity counseling	Physical activity counseling		
Outcomes assessment		Program Quality	Added the concept of program level outcomes assessment beyond simply patient level outcomes assessment to reflect initiatives since 2007 by Million Hearts, AACVPR, ACC, and AHA to increase enrollment and adherence, decrease the massive underutilization, and improve the equitable access of cardiac rehabilitation among all eligible participants	ACC/AHA Performance Measures7 Million Hearts Cardiac Rehabilitation Collaborative[26] AACVPR Performance Measures[27]

AACVPR indicates American Association of Cardiovascular and Pulmonary Rehabilitation; ACC; American College of Cariology; ADA, American Diabetes Association; AHA, American Heart Association; ASCVD, atherosclerotic cardiovascular disease; ATS, American Thoracic Society; and SSA, Social Security Act.

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to assess strength and balance even in someone who is functionally limited. The Short Physical Performance couples the 5 Times Sit-to-Stand with assessments of 4-meter gait speed and balance to achieve a more comprehensive composite of rudimentary function.[32]

Basic capacities of daily life can also be applied as primary metrics of function. Activities of daily living[33] are assessed as the simple count of difficulties in performing (or inability to perform) 5 activities: getting across a room, dressing, bathing or showering, eating, and getting in or out of bed. Instrumental activities of daily living are assessed as the simple count of difficulties in performing (or inability to perform) 5 activities: using a telephone, taking medication, handling money, shopping, and preparing meals.[33]

As part of initial assessments, it is important to clarify details of a patient's home environment (Table 2). This informs considerations to best achieve and sustain lifestyle goals and to also ascertain challenges pertaining to remote-based CR options. The logistics required for a patient to travel to site-based CR must be clarified, to determine whether barriers exist or there are scheduling options that should be prioritized. Other important considerations include whether the patient has broadband Internet access; is comfortable using a smartphone, computer, and wearables; and has access to community resources (eg, gym, community center, parks) that provide local options for exercise and physical activity.

An ITP provides a standardized approach to document each patient's assessment information, status, and goals of care. The ITP outlines intervention strategies for recovery and risk reduction and clarifies objectives to achieve long-term wellness plans. The ITP serves as a basis to communicate coherent and consistent treatment and follow-up plans with the patient and appropriate family members or domestic partners and the primary health care professionals. As part of this holistic therapeutic approach, the patient must be referred to appropriate specialists if indicated (eg, behavioral health specialist, mental health specialist, dietitian, medical consult, tobacco treatment, auditory evaluation). A final outcomes report is formulated to document evidence that a patient achieved CR core components of care as well as their personal goals of care. These final assessments include perspectives regarding the patient's success in taking appropriate doses of goal-directed medical therapy, receipt of vaccines pertinent to most cardiac patients (particularly influenza and pneumonia), a summary of long-term goals and strategies, and identification of areas that require further intervention and monitoring.

NUTRITIONAL COUNSELING

Dietary intake assessment is best achieved by food frequency questionnaires that are validated according to accepted standards[34] in CR populations, or those with similar demographic characteristics,[35] and reflect current guidelines for cardiovascular health.[36] Food frequency questionnaires yield a standardized score that reflects consumption of targeted nutrients or patterns of overall diet quality.[35] Documentation in the ITP should include diet scores and indices; a summary of eating habits and patterns, including behaviors to be targeted; and relevant clinical measures and comorbidities.[37]

Overall dietary recommendations follow patterns,[38] such as Dietary Approaches to Stop Hypertension, Mediterranean, US Department of Agriculture, vegetarian, or plant-based guidelines.[36,39,40] These plans emphasize positive means to consume healthful foods[41] and behaviors that facilitate limits on intake of saturated fats, sodium, sugars, and excess portions.[42] The individualized diet plan incorporates overall goals and SMART (specific, measurable, attainable and achievable, relevant, and time-bound) behaviors prioritizing the greatest aberrations from target revealed by the diet assessment,[36] identifies barriers and facilitators of change,[43] and is documented in the ITP. Patient education should include the rationale for the diet plan and the behavioral means to achieve it. Education can best be provided using various modalities, individually or in groups, in person or remotely,[13] and could use oral, written, video, digital, interactive, demonstration, or other modalities.[36,44] All CR staff provide general nutrition and behavioral guidance, although medical nutrition therapy can only be provided by a dietitian.[35]

Criteria for dietitian referral is based on clinical need, client request, and behavior change history. Dietitian engagement is more effective after 3 encounters[35] and is associated with decreased use of medications for diabetes, blood pressure, and lipids and improved effectiveness of medical treatment.[45] Dietitian engagement improves clinical outcomes in patients who have CVD,[44] newly diagnosed or uncontrolled diabetes,[44] heart failure,[46] malnourishment, or frailty, and in those who have not responded positively to previous dietary guidance.[39]

WEIGHT MANAGEMENT AND BODY COMPOSITION

Body fat to lean mass ratio (ie, body composition) is associated with CVD risk more closely than are weight and body mass index,[47–49] guide CR treatment plans, and can be used in the assessment of CR effectiveness. Individuals with a higher level of body fat and lower level of lean mass ("normal-weight obesity") are at particularly high CVD risk,[50] whereas those with a lower body fat level and higher lean mass level have lower CVD risk.[51] Improvements in body composition are associated with improved outcomes.[51] Prevalence of sarcopenia (increased muscle atrophy with

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decreased muscle strength or dynapenia) is highly prevalent in adults with CVD and contributes to frailty and cardiovascular risk.[52]

Waist circumference and waist-to-hip ratio are simple, feasible ways to assess body composition that correlate moderately well with body fat levels.[53,54] Bioelectrical impedance and water or air displacement are moderately accurate, but do not differentiate regional fat distribution, and the latter is limited by cost and complexity.[55] Dual X-ray absorptiometry and magnetic resonance imaging are highly accurate for regional and total fat mass and fat-free mass assessment but are limited by cost and complexity.[56] Additional studies of body composition assessment are needed in diverse racial and ethnic groups.[57] Waist circumference offers the most feasible and reasonably accurate method to assess body composition in remote CR care, assuming patients are able to perform measurements accurately. If available, bioelectrical impedance devices may be considered if patients are able to follow standard measurement protocols to limit the effect of variations in fluid volume on measurement accuracy. More advanced methods, such as dual X-ray absorptiometry or magnetic resonance imaging, are not feasible in remote settings. New and emerging digital tools may be helpful in assessing body composition remotely.[58]

Standard CR programs have limited effects in reducing weight and improving body composition. Highly focused and effective weight loss strategies within CR programs are needed to produce these improvements.[59] That being said, lifestyle habits help optimize body composition.[60] Aerobic exercise training helps increase lean mass and decrease body fat levels,[61] particularly high-intensity interval training.[62] Strength training is also essential to increasing lean mass.[63] A dietary pattern that optimizes calories, protein, complex carbohydrates, and nonsaturated fats can also help optimize body composition. Weight loss medications[64] and bariatric surgery[65] can help reduce body fat but may also reduce lean mass even when combined with exercise training.[66]

CVD AND RISK FACTOR MANAGEMENT

Comprehensive CVD education and risk factor management including health factors (eg, blood pressure, lipid levels, diabetes, weight) and health behaviors (eg, physical activity, diet, sleep, avoidance or cessation of unhealthful behaviors [eg, vaping, smoking] and substances [eg, alcohol, tobacco, marijuana, cocaine, opioids]) are the cornerstone of a multidisciplinary CR program.[7,19,67] Clinicians working in CR programs should educate patients about their disease process as well as symptom recognition and monitoring. In addition, it is important to engage patients in goal setting and action planning to foster enhanced patient activation and disease self-management. Advising patients on strategies to search for and identify trusted and reliable health information (eg, online and print resources from medical societies or medical centers) is imperative. Patient education is associated with a wide range of health benefits, including increases in disease knowledge, self-efficacy, tobacco cessation, medication adherence, physical activity, and healthful dietary behaviors.[68,69] These educational interventions can be delivered virtually while maintaining efficacy similar to that of education delivered in person.[70,71]

To maximize sustained clinical benefits for patients in CR, the evaluation of modifiable risk factors and implementation of lifestyle and pharmacologic interventions are paramount. The development of individual, patient-centered plans to optimize risk factors according to clinical practice guidelines is essential and should occur at initiation of CR, with periodic monitoring and refinement to support attainment of goals.[72]

Cigarette smoking is the leading cause of preventable death in the United States.[73,74] This behavior, and similar behaviors (eg, e-cigarette, pipe, cigar, hookah, marijuana use) should be addressed while the patient is in CR. Smoking cessation is a powerful intervention to reduce risk of recurrent heart disease and should be advised to all smokers.[75] Best practices for smoking cessation are described in societal guidelines.[25,76] When possible, it is helpful to have staff members gain additional expertise in smoking cessation counseling, pharmacotherapy, and motivational interviewing. Measurement of exhaled carbon monoxide may be useful for assessment and motivating patients to quit permanently.[77]

Specific recommendations for evaluation and interventions for risk factor management along with expected outcomes are listed in Table 2. Applicable clinical practice guidelines, performance measures, and other scientific statements that provide the support for these recommendations are summarized in the Supplemental Table.

PSYCHOSOCIAL MANAGEMENT

Psychological distress, especially depression, portends an elevated risk of adverse CVD outcomes,[78–81] including lower CR completion.[82] Other manifestations of distress (eg, perceived stress, anxiety) and psychosocial considerations (eg, sexual dysfunction, anger, hostility, loneliness, social isolation, substance use patterns that interfere with CR) influence CVD outcomes,[80] CR completion,[83] and cardiovascular recovery.[80,84] Thus, assessment of psychological distress and other psychosocial concerns warrant evaluation in CR. Increasing evidence highlights interactions between sleep[85,86] and anxiety sensitivity[87] with cardiovascular health. Future research is needed to determine whether assessing and targeting sleep and anxiety sensitivity benefits CR outcomes.

Psychosocial interventions with psychosocial risk factor education, coping skills training, and engagement in

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active therapeutic techniques (eg, cognitive behavioral therapy, stress management, relaxation) can improve psychological distress in CVD.[80,88] Previous research has primarily evaluated face-to-face psychosocial interventions in CVD. Some, but not all, evidence indicates that home-, Internet- and telephone-delivered interventions can address psychosocial concerns,[88,89] and that mobile health interventions support health behavior change.[89] Additional studies are needed to clarify how to reach diverse populations, which interventions are most potent, and the nature and effectiveness of remotely delivered psychosocial care.

Curating a supportive CR environment and leveraging community resources are affirmed by CR patient-reported benefits of social support[90,91] and some evidence of beneficial peer-supported interventions for varied health promotion outcomes.[92] However, prospective, randomized evidence of peer support interventions benefiting CR outcomes is lacking. Evidence gaps exist regarding how effectively remote delivery interventions achieve a supportive CR environment and whether reduced face-to-face peer and health care professional support affects psychosocial outcomes.

Persisting depression at CR completion confers increased mortality risk after CR, especially with concurrent anxiety and hostility.[93] Thus, the absence of clinically elevated psychosocial concerns at CR completion is desired. See Table 2 for full details of the assessment, intervention, and desired outcomes.

AEROBIC EXERCISE TRAINING

Aerobic exercise supplemented with resistance training remains the cornerstone intervention for CR. Although there are numerous benefits of exercise, including weight loss and reduction of CVD risk factors, improvement of cardiorespiratory fitness is of utmost importance and should therefore be assessed.[29,94] Prescribing aerobic exercise with the main goal of increasing cardiorespiratory fitness involves 4 components: frequency, intensity, duration, and mode of activity. The exercise volume, or sum of these components, can be expressed as the accumulated metabolic equivalents of task per week or calories burned in response to exercise per week. To help individualize the exercise intensity prescription, guidelines recommend performing graded exercise testing to objectively quantify peak oxygen uptake and target heart rate ranges, and account for ischemic thresholds, intensities associated with an abnormal blood pressure response, desaturation, symptoms, or arrhythmias (Table 2).[95] When graded exercise testing is not available, the Talk Test[96] or information gathered from a 6-minute walk test[97] may be used to help identify the appropriate exercise intensity[98] that promotes improvements in cardiorespiratory fitness. However, a patient's cardiorespiratory fitness may be underestimated due to the self-paced nature of a 6-minute walk test. The Borg Scale of Perceived Exertion (6–20 scale: moderate 12–13 and vigorous 14–16; 0–10 scale: moderate 3–4 and vigorous 5–7) can be used to interpret the patient's effort.[99]

CR programs commonly offer 3 sessions per week, which generally facilitates improvements in cardiorespiratory fitness; however, participants who are stable and capable should be encouraged to perform supplemental home exercise on non-CR days with the goal of exercising ≥5 days per week to meet federal physical activity guidelines.[100] In addition, patients who are stable and capable should perform aerobic exercise of moderate to vigorous intensity (Table 2) relative to their assessed cardiorespiratory fitness. Practitioners should avoid using resting heart rate plus 20 to 30 beats per minute or age-predicted maximal heart rate equations to calculate exercise intensities because of wide discrepancies between age-predicted and measured maximal heart rate.[101] For patients with sufficient stamina, exercise sessions can consist of 20 to 60 minutes of aerobic exercise, with the duration increased by 1 to 5 minutes per session until the exercise volume goal has been achieved.[102] The wide time range accounts for the intensity of exercise being performed (ie, less time is required at higher intensities to match the volume of activity performed at lower intensities for a longer time), incorporation of other exercises (eg, resistance, flexibility) within the allotted session time, as well as the patient's starting endurance. Aerobic exercise may be continuous or segmented with passive or active rest periods in between periods of activity at the prescribed intensity. For instance, high-intensity interval training has been implemented safely in CR programs, and has been noted to improve cardiorespiratory fitness significantly compared with moderate-intensity continuous training.[103] Detailed reviews on the practical implementation of high-intensity interval training can be found in other publications.[104,105] Common modes of aerobic exercise include walking, cycling, rowing, swimming, jogging, and elliptical trainers. The patient's preferences and physical abilities should be considered when selecting modes of activity. Integrating walking is of great importance, because this is the main form of physical activity performed throughout the day. Other modalities should also be encouraged, particularly modes that incorporate upper-extremity exercise as tolerated by individuals recovering from a median sternotomy in addition to patients with general upper-extremity weakness. These exercise considerations should ultimately support efforts to progress patients and have them meet the AACVPR functional capacity performance measure goals (Table 2).[106] Continuous electrocardiographic telemetry and exercise blood pressure monitoring likely add value to a CR program, especially in high-risk or symptomatic patients. However, exercise is safe in general,[107,108] and evidence is insufficient to determine how to best implement these services, how long to continue them, or which patients are most likely to benefit. Until

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such evidence is available, programs should provide these services as determined by insurance, regulatory requirements, or by clinical need as determined by the program medical director.

Newer delivery strategies have been deployed to engage individuals in exercise. Expanding beyond center-based CR, virtual and remote delivery of CR have emerged. Often, these programs rely on remote counseling with indirect supervision. Evidence suggests that these models may be a suitable alternative for low- to moderate-risk patients with CVD who lack access to center-based CR programs.[13] Similar improvements from exercise are seen with virtual and remote delivery of CR and may enable more patients to engage in exercise.[17] However, future studies are needed to investigate the delivery of virtual or remote CR to underserved communities facing environmental, financial, or technologic barriers.

STRENGTH TRAINING

Progressive resistance exercise training independently leads to improved strength, balance, metabolism, and functional independence in patients with CVD as well as increased cardiorespiratory fitness in low-functioning individuals.[109–112] Resistance training has value to all patients receiving CR but is often particularly important to build requisite capacity to tolerate aerobic exercise in patients who start CR relatively encumbered by deconditioning, frailty, or sarcopenia. To optimize outcomes, the resistance exercise training prescription should be customized to the patient's needs.[113] Performing strength, functional, and fall risk assessments at the entry or early phases of CR are encouraged, and are delineated in Table 2.

Resistance exercise training should initially be performed on 2 nonconsecutive days per week and progressed to 3 days per week. Whereas the gold standard method of prescribing resistance exercise intensity is on the basis of 1-repetition maximum testing (Table 2), patient hesitancy and lack of resistance machines or equipment serve as common barriers. The Borg scale may be used to select a resistance that elicits an initial perceived exertion of 11 to 14 (fairly light to somewhat hard).[99] Patients may start with 1 set per exercise and progress to 3 sets as tolerated with 10 to 15 repetitions per set while maintaining appropriate form and range of motion. Weight machines, free weights, elastic bands, or body weight can be used to elicit a stimulus that facilitates strength gains across populations.[114] When the upper repetition limit is achieved comfortably, the weight should be increased by ≈5%, or a resistance band with greater tension used. Eight to 10 exercises targeting major muscle groups should be performed in a session. When center-based CR session time constraints limit the number of exercises that can be completed, multijoint exercises (ie, row, chest press) can be emphasized. Resistance exercise training goals will differ relative to the patient's needs and desires. For older adults, preventing or treating frailty should be of high priority; resistance training and nutrition are both fundamental. Additional information on measures of physical frailty and function and appropriate interventions are discussed in a previous publication.[30]

Preliminary evidence supports the efficacy of virtual or remote resistance exercise training.[115] Due to comparatively less monitoring during remote versus in-person sessions, facilitating the patient's mastery of appropriate exercise form and breathing technique (out during contraction and in during relaxation) are crucial for maintaining patient safety. Future studies are needed to determine the feasibility and efficacy of screening patients virtually for frailty and delivering remote resistance exercise training interventions aimed at reversing frailty.

PHYSICAL ACTIVITY COUNSELING

Lack of physical activity is an independent risk factor for the development and progression of coronary disease[116]; therefore, a primary goal of CR is to increase total physical activity and help establish habitual routines to improve long-term adherence and reduce the risk of future cardiac events. Assessment of physical activity can be obtained through self-report with use of questionnaires or patient-recorded exercise diary, or objectively with physical activity monitors.[117]

Aerobic physical activity guidelines recommend ≥150 min/wk of moderate-intensity activity or 75 minutes of vigorous activity for substantial health benefits,[100] but a large proportion of patients receiving CR do not achieve these weekly targets.[118,119] This is pertinent given that sedentary time is associated with an increased risk of traditional CVD risk factors as well as mortality. Health benefits may still be acquired at levels well below the recommendations, however[100]; therefore, an initial emphasis on moving more and sitting less[120] can help with progressing patients toward meeting the physical activity guidelines. CR programs should include specific education about minimizing sedentary time and work with patients on ways to increase physical activity in their daily routine.

Pedometers and activity monitors have become popular tools for monitoring and promoting physical activity at a level that leads to improvements in cardiorespiratory fitness.[121] After a recent cardiac event, individuals have low physical activity levels, often <5000 steps per day.[116,122] Whereas a goal of 10 000 steps per day is often recommended, there is limited evidence regarding the specificity of this goal.[123,124] In a recent study of 4840 patients, a greater number of steps per day was associated with lower all-cause mortality risk, with an adjusted hazard ratio (HR) for 8000 steps per day versus 4000 steps per day of 0.49 (95% CI, 0.44–0.55).[125] In a large meta-analysis of 226 889 patients, a 1000-step increment was associated with a 15%

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reduced risk of all-cause mortality (HR, 0.85 [95% CI, 0.81–0.91]; $P<0.001$).[126] As opposed to time-based activity recommendations, step-count feedback can be a novel, low-cost approach to monitor and increase activity levels.[127] In addition, activity monitoring tools coupled with communication platforms (ie, visual, audio, text) that allow for remote interactions with clinicians working in CR programs have been shown to promote improvements in cardiorespiratory fitness and physical activity levels.[121,128,129] However, considerable variability in physical activity measurement and intervention outcomes across studies warrants future efforts to improve upon current applications of activity monitors in CR.

PROGRAM QUALITY

All CR programs should assess program quality comprehensively on at least an annual basis. This includes measuring quality at the system level (hospital and clinic referral), program level (enrollment, adherence, completion), and patient level (eg, improvement in fitness, risk factors).

Improving participation rates in CR is a national priority. All CR programs should work to solve this persistent problem. The Centers for Disease Control and Prevention supports the ongoing Million Hearts Cardiac Rehabilitation Collaborative, which aims to increase CR enrollment to 70% of all eligible patients.[26] In addition, the National Institutes of Health, the Agency for Healthcare Quality, and AACVPR have created separate initiatives to further enhance CR participation (Supplemental Table). Hybrid programming, including a combination of in-person, virtual, and remote sessions, is likely to be a key strategy to increase national CR uptake.[13]

The American College of Cardiology and American Heart Association have jointly published performance measures for use by US hospitals and health systems.[7] These measures are used as part of the Healthcare Effectiveness Data and Information Set to help hospitals and health systems improve CR participation rates. Local CR programs should partner with local hospitals and health systems to ensure that all eligible patients are referred and enrolled in CR. At the end of the program, a final outcomes report should be sent to the referring health care professional, detailing each patient's individualized assessments, interventions, improvement, and measured outcomes.

At the program level, AACVPR developed performance measures to assess enrollment and adherence.[27] At the patient level, AACVPR has established 4 performance measures: improvement in functional capacity, improvement in depression, blood pressure control, and tobacco intervention.[106] These measures should help programs assess local CR participation rates and program quality, compared with national metrics, and are part of the triennial AACVPR program certification applications.

Programs that carefully assess program quality will likely discover multiple performance gaps in the delivery of CR to their populations and patients (Supplemental Table). When problems are discovered, quality improvement efforts should be undertaken to reduce or eliminate programmatic quality gaps.

The exact nature of quality improvement efforts will depend on local circumstances and resources, but should ensure that underserved population needs are actively addressed, targeting widespread barriers associated with race, sex, age, frailty, rurality, comorbidity, and socioeconomics. Future research aimed at increasing CR participation should focus on including diverse populations, ensuring equitable delivery, and optimizing delivery models of care (in-person, virtual, remote, and hybrid) to maximize the benefits of CR to all eligible patients.

Given the urgent need for program assessment and quality improvement in CR, we consider program quality to be a core component of CR, regardless of delivery method (in person, virtual, or remote). Multiple local successful quality improvement projects have been undertaken and published, including dozens that are suggested in the Million Hearts Cardiac Rehabilitation Change Package.[130] AACVPR educational resources can help provide quality improvement guidance and direction. Programs are encouraged to review these previous projects and to improve local CR quality immediately.

CONCLUSIONS

CR is a high-value but greatly underused CVD intervention. High-quality CR programs that adhere to these core components and incorporate virtual and remote delivery are critical to increase enrollment in and adherence to CR and to reach traditionally underserved populations.

ARTICLE INFORMATION

The American Heart Association makes every effort to avoid any actual or potential conflicts of interest that may arise as a result of an outside relationship or a personal, professional, or business interest of a member of the writing panel. Specifically, all members of the writing group are required to complete and submit a Disclosure Questionnaire showing all such relationships that might be perceived as real or potential conflicts of interest.

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Writing Group Disclosures

Writing group member	Employment	Research grant	Other research support	Speakers' bureau/ honoraria	Expert witness	Ownership interest	Consultant/ advisory board	Other
Todd M. Brown	University of Alabama at Birmingham	None	None	None	None	None	None	None
Quinn R. Pack	Baystate Medical Center	None	None	None	None	None	None	None
Ellen Aberegg	Ellen Aberegg Consulting	None	None	None	None	None	None	None
LaPrincess C. Brewer	Mayo Clinic College of Medicine	Patient-Centered Outcomes Research Institute (comparative effectiveness of in-person and telehealth cardiac rehabilitation)†; National Institute of Diabetes and Digestive and Kidney Diseases (Creating Opportunities for Underrepresented Researchers to Achieve Growth and Excellence [COURAGE])†; Miami Heart Research Institute (Fostering African-American Improvement in Total Health [FAITH!])†; AHA - National (Virtual World-based Cardiac Rehabilitation to Improve Cardiovascular Health and Outcomes Among Cardiac Patients)†; Bristol-Myers Squibb Foundation, Inc (Virtual World-based Cardiac Rehabilitation to Improve Cardiovascular Health and Outcomes Among Cardiac Patients: A Multicenter Randomized Clinical Trial)†; National Institute on Minority Health and Health Disparities (Techquity by FAITH!: a cluster randomized controlled trial to assess the efficacy of a community-informed, CV health promotion mobile health intervention w/digital health advocate support [supplement])†	None	None	None	None	None	None
Yvonne R. Ford	North Carolina A&T State University School of Nursing	None	None	None	None	None	None	None
Daniel E. Forman	University of Pittsburgh Medical Center	None	None	None	None	None	None	None
Emily C. Gathright	The Miriam Hospital and Brown University	None	None	None	None	None	None	None
Sherrie Khadanga	University of Vermont Medical Center	NIA (just received an R01)*	None	None	None	None	None	None
Cemal Ozemek	University of Illinois at Chicago	None	None	None	None	None	None	None
Randal J. Thomas	Mayo Clinic	None	None	None	None	None	None	None

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Alison L. Bailey	HCA Healthcare (Chattanooga, TN)	None	None	None	None	None	Pritikin*	None
Theresa M. Beckie	University of South Florida	None	None	None	None	None	None	None
Eileen Collins	University of Illinois at Chicago College of Nursing	None	None	None	None	None	None	None
John W. Hughes	Kent State University	Co-PI (PI – Cynthia Osborn)†	None	None	None	None	Consultant (PI – Park)†	None
Richard A. Josephson	Retired	None	None	None	None	None	None	None
Leonard A. Kaminsky	Ball State University	None	None	None	None	None	None	None
Steven J. Keteyian	Henry Ford Medical Group	NHLBI (study of hybrid cardiac rehabilitation)†	None	None	None	Nimble Heart*	Abt Associates†; Kento Health†	None
Carl "Chip" Lavie	John Ochsner Heart and Vascular Institute	None	None	None	None	None	None	None
Katherine Menson	University of Vermont	None	None	None	None	None	None	None
Jonathan Myers	VA Palo Alto Healthcare System and Stanford University	None	None	None	None	None	None	None
Paul I. Oh	Toronto Rehabilitation Institute/ University Health Network (Canada)	None	None	None	None	None	Novartis*	None
David Prince	Montefiore Medical Center	None	None	None	None	None	None	None

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